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Naïve Bayes Classifier

A Person	Documents
Training	- He is a bad man - He is a liar - He is not helpful + He is an educated person + He loves animals very much + He achieved 2 medals from the university.
Test	? Very helpful to animals

The prior $P(c)$ for 2 classes is computed as $\frac{N_c}{N_{doc}}$ or N_c / N_{doc} :

$$P(-) = \frac{3}{6}, \quad P(+)=\frac{3}{6}$$

The likelihood from the training set for the remaining 3 words "very", "helpful" & "animal" are as follows:-

$$P(\text{"very"}|-) = \frac{0+1}{13+21} = \frac{1}{34}$$

$$P(\text{"helpful"}|-) = \frac{1+1}{13+21} = \frac{2}{34}$$

$$P(\text{"animals"}|-) = \frac{0+1}{13+21} = \frac{1}{34}$$

$$P(\text{"very"}|+) = \frac{1+1}{17+21} = \frac{2}{38}$$

$$P(\text{"helpful"}|+) = \frac{0+1}{17+21} = \frac{1}{38}$$

$$P(\text{"animals"}|+) = \frac{1+1}{17+21} = \frac{2}{38}$$

For the test sentence $S = \text{"very helpful to animals"}$ after removing the word "to" the chosen class is therefore computed as follows:-

$$P(-) P(S|-) = \frac{3}{6} \times \frac{1 \times 2 \times 1}{34^3} = 2.54 \times 10^{-5}$$

$$P(+) P(S|+) = \frac{3}{6} \times \frac{2 \times 1 \times 2}{38^3} = 3.64 \times 10^{-5}$$

The model thus predicts the class positive for the test sentence.